

Retrieval-Augmented Generation (RAG) and Artificial Intelligence

What is AI?

Artificial Intelligence (AI) is the field of computer science focused on creating systems that perform tasks requiring human intelligence. These tasks include reasoning, learning, perception, language understanding, and decision-making. Modern AI uses machine learning and deep learning to identify patterns in large datasets.

What is RAG?

Retrieval-Augmented Generation (RAG) combines a retrieval system with a large language model. Instead of relying only on the model's internal knowledge, RAG searches external documents, retrieves the most relevant passages, and uses them to generate grounded answers. This reduces hallucinations and enables answers based on up-to-date or private knowledge.

RAG Pipeline

A typical RAG pipeline consists of document ingestion, text chunking, embedding generation, vector database indexing, similarity search, prompt construction, and response generation. Popular vector databases include FAISS, ChromaDB, Pinecone, and Milvus. Embedding models convert text into numerical vectors that capture semantic meaning.

Applications

RAG is widely used for enterprise knowledge assistants, document question answering, customer support bots, legal research, healthcare information systems, and educational assistants.

Best Practices

Good chunk sizes, overlap between chunks, metadata filtering, reranking, prompt engineering, and evaluation metrics such as precision and recall improve RAG quality. Security measures should protect sensitive documents and user privacy.

Sample Questions for Testing

1. What is Retrieval-Augmented Generation? 2. Why does RAG reduce hallucinations? 3. Name two vector databases. 4. What are embeddings? 5. List three applications of RAG. 6. What are the main stages of a RAG pipeline?